

project_4b4b

December 7, 2025

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[2]: import argparse
import os
import time
import shutil

import torch
import torch.nn as nn
import torch.optim as optim
import torch.nn.functional as F
import torch.backends.cudnn as cudnn

import torchvision
import torchvision.transforms as transforms

from models import *

global best_prec
use_gpu = torch.cuda.is_available()
print('=> Building model...')

batch_size = 128
model_name = "VGG16_project"
model = VGG16_project()

print(model)

normalize = transforms.Normalize(mean=[0.491, 0.482, 0.447], std=[0.247, 0.243,
↪0.262])

train_dataset = torchvision.datasets.CIFAR10(
    root='./data',
    train=True,
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download=True,
transform=transforms.Compose([
    transforms.RandomCrop(32, padding=4),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    normalize,
]))
trainloader = torch.utils.data.DataLoader(train_dataset, batch_size=batch_size,
↳shuffle=True, num_workers=2)

test_dataset = torchvision.datasets.CIFAR10(
    root='./data',
    train=False,
    download=True,
    transform=transforms.Compose([
        transforms.ToTensor(),
        normalize,
    ]))

testloader = torch.utils.data.DataLoader(test_dataset, batch_size=batch_size,
↳shuffle=False, num_workers=2)

print_freq = 100 # every 100 batches, accuracy printed. Here, each batch
↳includes "batch_size" data points
# CIFAR10 has 50,000 training data, and 10,000 validation data.

def train(trainloader, model, criterion, optimizer, epoch):
    batch_time = AverageMeter()
    data_time = AverageMeter()
    losses = AverageMeter()
    top1 = AverageMeter()

    model.train()

    end = time.time()
    for i, (input, target) in enumerate(trainloader):
        # measure data loading time
        data_time.update(time.time() - end)

        input, target = input.cuda(), target.cuda()

        # compute output
        output = model(input)
        loss = criterion(output, target)

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        # measure accuracy and record loss
        prec = accuracy(output, target)[0]
        losses.update(loss.item(), input.size(0))
        top1.update(prec.item(), input.size(0))

        # compute gradient and do SGD step
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        # measure elapsed time
        batch_time.update(time.time() - end)
        end = time.time()

    if i % print_freq == 0:
        print('Epoch: [{0}] [{1}/{2}]\t'
              'Time {batch_time.val:.3f} ({batch_time.avg:.3f})\t'
              'Data {data_time.val:.3f} ({data_time.avg:.3f})\t'
              'Loss {loss.val:.4f} ({loss.avg:.4f})\t'
              'Prec {top1.val:.3f}% ({top1.avg:.3f}%)'.format(
                  epoch, i, len(trainloader), batch_time=batch_time,
                  data_time=data_time, loss=losses, top1=top1))

def validate(val_loader, model, criterion ):
    batch_time = AverageMeter()
    losses = AverageMeter()
    top1 = AverageMeter()

    # switch to evaluate mode
    model.eval()

    end = time.time()
    with torch.no_grad():
        for i, (input, target) in enumerate(val_loader):

            input, target = input.cuda(), target.cuda()

            # compute output
            output = model(input)
            loss = criterion(output, target)

            # measure accuracy and record loss
            prec = accuracy(output, target)[0]
            losses.update(loss.item(), input.size(0))

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        top1.update(prec.item(), input.size(0))

        # measure elapsed time
        batch_time.update(time.time() - end)
        end = time.time()

        if i % print_freq == 0: # This line shows how frequently print out
            ↪ the status. e.g., i%5 => every 5 batch, prints out
                print('Test: [{0}/{1}]\t'
                      'Time {batch_time.val:.3f} ({batch_time.avg:.3f})\t'
                      'Loss {loss.val:.4f} ({loss.avg:.4f})\t'
                      'Prec {top1.val:.3f}% ({top1.avg:.3f}%)'.format(
                        i, len(val_loader), batch_time=batch_time, loss=losses,
                        top1=top1))

    print(' * Prec {top1.avg:.3f}% '.format(top1=top1))
    return top1.avg

def accuracy(output, target, topk=(1,)):
    """Computes the precision@k for the specified values of k"""
    maxk = max(topk)
    batch_size = target.size(0)

    _, pred = output.topk(maxk, 1, True, True)
    pred = pred.t()
    correct = pred.eq(target.view(1, -1).expand_as(pred))

    res = []
    for k in topk:
        correct_k = correct[:k].view(-1).float().sum(0)
        res.append(correct_k.mul_(100.0 / batch_size))
    return res

class AverageMeter(object):
    """Computes and stores the average and current value"""
    def __init__(self):
        self.reset()

    def reset(self):
        self.val = 0
        self.avg = 0
        self.sum = 0
        self.count = 0

    def update(self, val, n=1):

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        self.val = val
        self.sum += val * n
        self.count += n
        self.avg = self.sum / self.count

def save_checkpoint(state, is_best, fdir):
    filepath = os.path.join(fdir, 'checkpoint.pth')
    torch.save(state, filepath)
    if is_best:
        shutil.copyfile(filepath, os.path.join(fdir, 'model_best.pth.tar'))

#model = nn.DataParallel(model).cuda()
#all_params = checkpoint['state_dict']
#model.load_state_dict(all_params, strict=False)
#criterion = nn.CrossEntropyLoss().cuda()
#validate(testloader, model, criterion)

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=> Building model...

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VGG_quant(
  (features): Sequential(
    (0): QuantConv2d(
      3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): QuantConv2d(
      64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
    (4): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (5): ReLU(inplace=True)
    (6): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (7): QuantConv2d(
      64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
    (8): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (9): ReLU(inplace=True)
    (10): QuantConv2d(
      128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
  )
)

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    )
    (11): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (12): ReLU(inplace=True)
    (13): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (14): QuantConv2d(
        128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (15): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (16): ReLU(inplace=True)
    (17): QuantConv2d(
        256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (18): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (19): ReLU(inplace=True)
    (20): QuantConv2d(
        256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (21): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (22): ReLU(inplace=True)
    (23): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (24): QuantConv2d(
        256, 8, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (25): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (26): ReLU(inplace=True)
    (27): QuantConv2d(
        8, 8, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (28): ReLU(inplace=True)
    (29): QuantConv2d(
        8, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (30): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (31): ReLU(inplace=True)

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(32): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
(33): QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
)
(34): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
(35): ReLU(inplace=True)
(36): QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
)
(37): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
(38): ReLU(inplace=True)
(39): QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
)
(40): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
(41): ReLU(inplace=True)
(42): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
(43): AvgPool2d(kernel_size=1, stride=1, padding=0)
)
(classifier): Linear(in_features=512, out_features=10, bias=True)
)
100%|      | 170M/170M [00:06<00:00, 27.3MB/s]

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[ ]: def adjust_learning_rate(optimizer, epoch):
    """For resnet, the lr starts from 0.1, and is divided by 10 at 80 and 120_
    epochs"""
    adjust_list = [60, 80]
    if epoch in adjust_list:
        for param_group in optimizer.param_groups:
            param_group['lr'] = param_group['lr'] * 0.1

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[ ]: # This cell won't be given, but students will complete the training

lr = 4e-2
weight_decay = 1e-4
epochs = 100
best_prec = 0

#model = nn.DataParallel(model).cuda()

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model.cuda()
criterion = nn.CrossEntropyLoss().cuda()
optimizer = torch.optim.SGD(model.parameters(), lr=lr, momentum=0.9,
    ↪weight_decay=weight_decay)
# cudnn.benchmark = True

if not os.path.exists('result'):
    os.makedirs('result')
fdir = 'result/'+str(model_name)
if not os.path.exists(fdir):
    os.makedirs(fdir)

for epoch in range(0, epochs):
    adjust_learning_rate(optimizer, epoch)

    train(trainloader, model, criterion, optimizer, epoch)

    # evaluate on test set
    print("Validation starts")
    prec = validate(testloader, model, criterion)

    # remember best precision and save checkpoint
    is_best = prec > best_prec
    best_prec = max(prec, best_prec)
    print('best acc: {:.1f}'.format(best_prec))
    save_checkpoint({
        'epoch': epoch + 1,
        'state_dict': model.state_dict(),
        'best_prec': best_prec,
        'optimizer': optimizer.state_dict(),
    }, is_best, fdir)

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Epoch: [0] [0/391]      Time 0.727 (0.727)      Data 0.114 (0.114)      Loss
2.7576 (2.7576)      Prec 6.250% (6.250%)
Epoch: [0] [100/391]    Time 0.023 (0.031)      Data 0.002 (0.004)      Loss
2.2631 (3.2371)      Prec 15.625% (11.239%)
Epoch: [0] [200/391]    Time 0.022 (0.028)      Data 0.001 (0.004)      Loss
2.2255 (2.7506)      Prec 10.938% (12.310%)
Epoch: [0] [300/391]    Time 0.023 (0.028)      Data 0.001 (0.004)      Loss
2.1637 (2.5610)      Prec 12.500% (13.489%)
Validation starts
Test: [0/79]      Time 0.106 (0.106)      Loss 1.9992 (1.9992)      Prec 25.781%
(25.781%)
* Prec 23.890%
best acc: 23.890000
Epoch: [1] [0/391]      Time 0.159 (0.159)      Data 0.126 (0.126)      Loss

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2.0727 (2.0727)    Prec 18.750% (18.750%)
Epoch: [1][100/391]    Time 0.027 (0.026)    Data 0.005 (0.004)    Loss
1.9641 (2.0087)    Prec 18.750% (24.010%)
Epoch: [1][200/391]    Time 0.033 (0.026)    Data 0.007 (0.004)    Loss
1.8855 (1.9793)    Prec 25.000% (24.351%)
Epoch: [1][300/391]    Time 0.028 (0.026)    Data 0.006 (0.003)    Loss
1.8949 (1.9505)    Prec 28.125% (24.961%)
Validation starts
Test: [0/79]    Time 0.107 (0.107)    Loss 1.7862 (1.7862)    Prec 33.594%
(33.594%)
* Prec 31.710%
best acc: 31.710000
Epoch: [2][0/391]    Time 0.156 (0.156)    Data 0.126 (0.126)    Loss
1.7624 (1.7624)    Prec 27.344% (27.344%)
Epoch: [2][100/391]    Time 0.027 (0.026)    Data 0.005 (0.004)    Loss
1.6989 (1.8058)    Prec 33.594% (30.956%)
Epoch: [2][200/391]    Time 0.025 (0.025)    Data 0.002 (0.004)    Loss
1.8278 (1.7851)    Prec 32.031% (31.973%)
Epoch: [2][300/391]    Time 0.024 (0.026)    Data 0.002 (0.003)    Loss
1.6233 (1.7602)    Prec 43.750% (33.303%)
Validation starts
Test: [0/79]    Time 0.104 (0.104)    Loss 1.6709 (1.6709)    Prec 38.281%
(38.281%)
* Prec 36.610%
best acc: 36.610000
Epoch: [3][0/391]    Time 0.169 (0.169)    Data 0.132 (0.132)    Loss
1.7451 (1.7451)    Prec 39.844% (39.844%)
Epoch: [3][100/391]    Time 0.028 (0.027)    Data 0.006 (0.005)    Loss
1.5547 (1.6084)    Prec 48.438% (40.091%)
Epoch: [3][200/391]    Time 0.028 (0.026)    Data 0.006 (0.004)    Loss
1.6634 (1.5759)    Prec 42.969% (41.305%)
Epoch: [3][300/391]    Time 0.022 (0.026)    Data 0.001 (0.004)    Loss
1.6200 (1.5516)    Prec 40.625% (42.011%)
Validation starts
Test: [0/79]    Time 0.106 (0.106)    Loss 1.4282 (1.4282)    Prec 47.656%
(47.656%)
* Prec 44.980%
best acc: 44.980000
Epoch: [4][0/391]    Time 0.150 (0.150)    Data 0.121 (0.121)    Loss
1.4798 (1.4798)    Prec 43.750% (43.750%)
Epoch: [4][100/391]    Time 0.029 (0.026)    Data 0.008 (0.005)    Loss
1.4768 (1.4004)    Prec 44.531% (47.517%)
Epoch: [4][200/391]    Time 0.022 (0.026)    Data 0.001 (0.004)    Loss
1.3511 (1.3788)    Prec 55.469% (48.562%)
Epoch: [4][300/391]    Time 0.021 (0.026)    Data 0.001 (0.004)    Loss
1.4234 (1.3618)    Prec 43.750% (49.447%)
Validation starts
Test: [0/79]    Time 0.108 (0.108)    Loss 1.2444 (1.2444)    Prec 52.344%

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(52.344%)

* Prec 54.060%

best acc: 54.060000

Epoch: [5] [0/391]	Time 0.150 (0.150)	Data 0.121 (0.121)	Loss
1.2382 (1.2382)	Prec 52.344% (52.344%)		
Epoch: [5] [100/391]	Time 0.031 (0.026)	Data 0.010 (0.005)	Loss
1.0295 (1.2440)	Prec 67.188% (54.440%)		
Epoch: [5] [200/391]	Time 0.027 (0.026)	Data 0.004 (0.004)	Loss
1.2688 (1.2367)	Prec 53.125% (54.967%)		
Epoch: [5] [300/391]	Time 0.028 (0.026)	Data 0.006 (0.004)	Loss
1.2357 (1.2166)	Prec 54.688% (55.539%)		

Validation starts

Test: [0/79] Time 0.107 (0.107) Loss 1.0753 (1.0753) Prec 60.938% (60.938%)

* Prec 56.720%

best acc: 56.720000

Epoch: [6] [0/391]	Time 0.155 (0.155)	Data 0.126 (0.126)	Loss
1.0527 (1.0527)	Prec 59.375% (59.375%)		
Epoch: [6] [100/391]	Time 0.024 (0.026)	Data 0.003 (0.004)	Loss
1.0934 (1.0901)	Prec 57.812% (60.295%)		
Epoch: [6] [200/391]	Time 0.026 (0.025)	Data 0.002 (0.004)	Loss
0.9498 (1.0862)	Prec 65.625% (60.650%)		
Epoch: [6] [300/391]	Time 0.023 (0.025)	Data 0.002 (0.004)	Loss
1.1131 (1.0751)	Prec 61.719% (61.184%)		

Validation starts

Test: [0/79] Time 0.104 (0.104) Loss 0.9853 (0.9853) Prec 61.719% (61.719%)

* Prec 61.460%

best acc: 61.460000

Epoch: [7] [0/391]	Time 0.162 (0.162)	Data 0.125 (0.125)	Loss
0.9723 (0.9723)	Prec 64.062% (64.062%)		
Epoch: [7] [100/391]	Time 0.028 (0.026)	Data 0.005 (0.004)	Loss
0.8207 (0.9819)	Prec 73.438% (64.790%)		
Epoch: [7] [200/391]	Time 0.024 (0.026)	Data 0.002 (0.004)	Loss
0.9020 (0.9603)	Prec 64.844% (65.633%)		
Epoch: [7] [300/391]	Time 0.021 (0.026)	Data 0.001 (0.004)	Loss
0.8182 (0.9532)	Prec 77.344% (66.048%)		

Validation starts

Test: [0/79] Time 0.104 (0.104) Loss 1.0521 (1.0521) Prec 59.375% (59.375%)

* Prec 64.190%

best acc: 64.190000

Epoch: [8] [0/391]	Time 0.153 (0.153)	Data 0.125 (0.125)	Loss
0.7905 (0.7905)	Prec 71.875% (71.875%)		
Epoch: [8] [100/391]	Time 0.024 (0.026)	Data 0.002 (0.004)	Loss
0.7690 (0.8781)	Prec 68.750% (68.874%)		
Epoch: [8] [200/391]	Time 0.022 (0.025)	Data 0.001 (0.004)	Loss
0.8254 (0.8672)	Prec 64.844% (69.337%)		

Epoch: [8][300/391] Time 0.021 (0.025) Data 0.001 (0.004) Loss
0.7795 (0.8597) Prec 71.094% (69.557%)
Validation starts
Test: [0/79] Time 0.104 (0.104) Loss 0.8161 (0.8161) Prec 67.188%
(67.188%)
* Prec 67.420%
best acc: 67.420000
Epoch: [9][0/391] Time 0.158 (0.158) Data 0.121 (0.121) Loss
0.7810 (0.7810) Prec 71.875% (71.875%)
Epoch: [9][100/391] Time 0.026 (0.026) Data 0.005 (0.004) Loss
0.6293 (0.7995) Prec 78.125% (72.215%)
Epoch: [9][200/391] Time 0.021 (0.026) Data 0.000 (0.004) Loss
0.7613 (0.8011) Prec 73.438% (72.128%)
Epoch: [9][300/391] Time 0.024 (0.026) Data 0.002 (0.004) Loss
0.7916 (0.7972) Prec 73.438% (72.161%)
Validation starts
Test: [0/79] Time 0.105 (0.105) Loss 0.7109 (0.7109) Prec 73.438%
(73.438%)
* Prec 71.680%
best acc: 71.680000
Epoch: [10][0/391] Time 0.151 (0.151) Data 0.121 (0.121) Loss
0.6801 (0.6801) Prec 77.344% (77.344%)
Epoch: [10][100/391] Time 0.026 (0.026) Data 0.005 (0.004) Loss
0.7838 (0.7429) Prec 72.656% (73.731%)
Epoch: [10][200/391] Time 0.026 (0.025) Data 0.005 (0.004) Loss
0.8531 (0.7358) Prec 71.094% (74.168%)
Epoch: [10][300/391] Time 0.027 (0.025) Data 0.006 (0.004) Loss
0.4470 (0.7336) Prec 86.719% (74.434%)
Validation starts
Test: [0/79] Time 0.107 (0.107) Loss 0.7071 (0.7071) Prec 76.562%
(76.562%)
* Prec 73.370%
best acc: 73.370000
Epoch: [11][0/391] Time 0.158 (0.158) Data 0.129 (0.129) Loss
0.7250 (0.7250) Prec 72.656% (72.656%)
Epoch: [11][100/391] Time 0.028 (0.026) Data 0.006 (0.005) Loss
0.6505 (0.6908) Prec 79.688% (75.913%)
Epoch: [11][200/391] Time 0.026 (0.026) Data 0.004 (0.004) Loss
0.6905 (0.6846) Prec 71.875% (76.271%)
Epoch: [11][300/391] Time 0.024 (0.025) Data 0.002 (0.004) Loss
0.6696 (0.6816) Prec 72.656% (76.420%)
Validation starts
Test: [0/79] Time 0.107 (0.107) Loss 0.8356 (0.8356) Prec 73.438%
(73.438%)
* Prec 74.250%
best acc: 74.250000
Epoch: [12][0/391] Time 0.149 (0.149) Data 0.120 (0.120) Loss
0.6546 (0.6546) Prec 77.344% (77.344%)

Epoch: [12][100/391] Time 0.026 (0.026) Data 0.005 (0.004) Loss
0.6461 (0.6387) Prec 78.906% (77.924%)

Epoch: [12][200/391] Time 0.023 (0.025) Data 0.002 (0.003) Loss
0.7465 (0.6358) Prec 74.219% (78.133%)

Epoch: [12][300/391] Time 0.028 (0.025) Data 0.007 (0.003) Loss
0.5723 (0.6384) Prec 78.906% (78.086%)

Validation starts

Test: [0/79] Time 0.105 (0.105) Loss 0.6889 (0.6889) Prec 75.000%
(75.000%)

* Prec 77.690%

best acc: 77.690000

Epoch: [13][0/391] Time 0.148 (0.148) Data 0.119 (0.119) Loss
0.6529 (0.6529) Prec 82.031% (82.031%)

Epoch: [13][100/391] Time 0.028 (0.026) Data 0.007 (0.004) Loss
0.4339 (0.6006) Prec 87.500% (79.394%)

Epoch: [13][200/391] Time 0.031 (0.026) Data 0.008 (0.004) Loss
0.6827 (0.5992) Prec 77.344% (79.513%)

Epoch: [13][300/391] Time 0.029 (0.025) Data 0.006 (0.004) Loss
0.7290 (0.5963) Prec 75.781% (79.625%)

Validation starts

Test: [0/79] Time 0.097 (0.097) Loss 0.7076 (0.7076) Prec 74.219%
(74.219%)

* Prec 76.050%

best acc: 77.690000

Epoch: [14][0/391] Time 0.150 (0.150) Data 0.122 (0.122) Loss
0.5337 (0.5337) Prec 80.469% (80.469%)

Epoch: [14][100/391] Time 0.023 (0.026) Data 0.002 (0.005) Loss
0.5641 (0.5711) Prec 82.031% (80.569%)

Epoch: [14][200/391] Time 0.020 (0.025) Data 0.001 (0.005) Loss
0.5715 (0.5668) Prec 78.906% (80.772%)

Epoch: [14][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.5024 (0.5671) Prec 80.469% (80.824%)

Validation starts

Test: [0/79] Time 0.094 (0.094) Loss 0.7577 (0.7577) Prec 75.000%
(75.000%)

* Prec 74.810%

best acc: 77.690000

Epoch: [15][0/391] Time 0.153 (0.153) Data 0.118 (0.118) Loss
0.6014 (0.6014) Prec 78.906% (78.906%)

Epoch: [15][100/391] Time 0.027 (0.026) Data 0.005 (0.005) Loss
0.6038 (0.5401) Prec 77.344% (81.459%)

Epoch: [15][200/391] Time 0.029 (0.025) Data 0.006 (0.004) Loss
0.4504 (0.5388) Prec 85.938% (81.759%)

Epoch: [15][300/391] Time 0.029 (0.025) Data 0.007 (0.004) Loss
0.5739 (0.5345) Prec 82.031% (81.863%)

Validation starts

Test: [0/79] Time 0.107 (0.107) Loss 0.6842 (0.6842) Prec 76.562%
(76.562%)

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* Prec 78.450%
best acc: 78.450000
Epoch: [16][0/391]      Time 0.150 (0.150)      Data 0.121 (0.121)      Loss
0.4350 (0.4350)      Prec 85.156% (85.156%)
Epoch: [16][100/391]    Time 0.026 (0.026)      Data 0.004 (0.004)      Loss
0.5600 (0.5176)      Prec 80.469% (81.969%)
Epoch: [16][200/391]    Time 0.025 (0.025)      Data 0.002 (0.003)      Loss
0.3594 (0.5165)      Prec 86.719% (82.311%)
Epoch: [16][300/391]    Time 0.027 (0.025)      Data 0.006 (0.003)      Loss
0.5676 (0.5118)      Prec 78.125% (82.400%)
Validation starts
Test: [0/79]      Time 0.108 (0.108)      Loss 0.4903 (0.4903)      Prec 85.156%
(85.156%)
* Prec 80.930%
best acc: 80.930000
Epoch: [17][0/391]      Time 0.159 (0.159)      Data 0.121 (0.121)      Loss
0.4536 (0.4536)      Prec 85.156% (85.156%)
Epoch: [17][100/391]    Time 0.028 (0.027)      Data 0.006 (0.004)      Loss
0.5495 (0.4871)      Prec 81.250% (83.679%)
Epoch: [17][200/391]    Time 0.028 (0.026)      Data 0.007 (0.004)      Loss
0.6001 (0.4808)      Prec 78.125% (83.757%)
Epoch: [17][300/391]    Time 0.026 (0.026)      Data 0.004 (0.003)      Loss
0.5315 (0.4834)      Prec 84.375% (83.513%)
Validation starts
Test: [0/79]      Time 0.107 (0.107)      Loss 0.5575 (0.5575)      Prec 81.250%
(81.250%)
* Prec 78.610%
best acc: 80.930000
Epoch: [18][0/391]      Time 0.148 (0.148)      Data 0.119 (0.119)      Loss
0.4060 (0.4060)      Prec 86.719% (86.719%)
Epoch: [18][100/391]    Time 0.027 (0.026)      Data 0.005 (0.004)      Loss
0.3922 (0.4617)      Prec 85.938% (84.182%)
Epoch: [18][200/391]    Time 0.023 (0.025)      Data 0.002 (0.004)      Loss
0.4645 (0.4583)      Prec 85.938% (84.433%)
Epoch: [18][300/391]    Time 0.028 (0.025)      Data 0.006 (0.003)      Loss
0.3450 (0.4604)      Prec 89.062% (84.367%)
Validation starts
Test: [0/79]      Time 0.107 (0.107)      Loss 0.5035 (0.5035)      Prec 83.594%
(83.594%)
* Prec 82.100%
best acc: 82.100000
Epoch: [19][0/391]      Time 0.150 (0.150)      Data 0.122 (0.122)      Loss
0.4263 (0.4263)      Prec 87.500% (87.500%)
Epoch: [19][100/391]    Time 0.021 (0.026)      Data 0.001 (0.005)      Loss
0.5084 (0.4422)      Prec 85.938% (85.125%)
Epoch: [19][200/391]    Time 0.024 (0.025)      Data 0.002 (0.004)      Loss
0.5009 (0.4426)      Prec 85.156% (85.094%)
Epoch: [19][300/391]    Time 0.028 (0.025)      Data 0.006 (0.004)      Loss

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0.3793 (0.4411) Prec 89.062% (85.073%)
Validation starts
Test: [0/79] Time 0.108 (0.108) Loss 0.4473 (0.4473) Prec 84.375% (84.375%)
* Prec 81.420%
best acc: 82.100000
Epoch: [20][0/391] Time 0.152 (0.152) Data 0.123 (0.123) Loss 0.4421 (0.4421) Prec 82.812% (82.812%)
Epoch: [20][100/391] Time 0.024 (0.026) Data 0.002 (0.005) Loss 0.4731 (0.4193) Prec 82.031% (85.644%)
Epoch: [20][200/391] Time 0.023 (0.025) Data 0.002 (0.004) Loss 0.4679 (0.4228) Prec 82.031% (85.553%)
Epoch: [20][300/391] Time 0.027 (0.025) Data 0.006 (0.004) Loss 0.5220 (0.4245) Prec 82.031% (85.564%)
Validation starts
Test: [0/79] Time 0.107 (0.107) Loss 0.4024 (0.4024) Prec 85.938% (85.938%)
* Prec 82.460%
best acc: 82.460000
Epoch: [21][0/391] Time 0.148 (0.148) Data 0.120 (0.120) Loss 0.5299 (0.5299) Prec 84.375% (84.375%)
Epoch: [21][100/391] Time 0.026 (0.026) Data 0.004 (0.004) Loss 0.3735 (0.4028) Prec 86.719% (86.804%)
Epoch: [21][200/391] Time 0.025 (0.025) Data 0.004 (0.004) Loss 0.4031 (0.4087) Prec 89.062% (86.466%)
Epoch: [21][300/391] Time 0.021 (0.025) Data 0.001 (0.004) Loss 0.3520 (0.4075) Prec 89.062% (86.311%)
Validation starts
Test: [0/79] Time 0.104 (0.104) Loss 0.3897 (0.3897) Prec 85.938% (85.938%)
* Prec 83.170%
best acc: 83.170000
Epoch: [22][0/391] Time 0.154 (0.154) Data 0.125 (0.125) Loss 0.4098 (0.4098) Prec 85.156% (85.156%)
Epoch: [22][100/391] Time 0.023 (0.026) Data 0.002 (0.004) Loss 0.3602 (0.3733) Prec 88.281% (87.438%)
Epoch: [22][200/391] Time 0.021 (0.026) Data 0.001 (0.004) Loss 0.4156 (0.3818) Prec 86.719% (86.960%)
Epoch: [22][300/391] Time 0.022 (0.026) Data 0.001 (0.004) Loss 0.4418 (0.3819) Prec 82.812% (86.890%)
Validation starts
Test: [0/79] Time 0.105 (0.105) Loss 0.4660 (0.4660) Prec 84.375% (84.375%)
* Prec 81.540%
best acc: 83.170000
Epoch: [23][0/391] Time 0.152 (0.152) Data 0.123 (0.123) Loss 0.5323 (0.5323) Prec 83.594% (83.594%)
Epoch: [23][100/391] Time 0.028 (0.026) Data 0.006 (0.004) Loss

0.2898 (0.3590) Prec 88.281% (87.864%)
 Epoch: [23][200/391] Time 0.028 (0.025) Data 0.006 (0.004) Loss
 0.2956 (0.3655) Prec 91.406% (87.582%)
 Epoch: [23][300/391] Time 0.024 (0.025) Data 0.003 (0.003) Loss
 0.3378 (0.3720) Prec 87.500% (87.407%)
 Validation starts
 Test: [0/79] Time 0.105 (0.105) Loss 0.4018 (0.4018) Prec 89.062%
 (89.062%)
 * Prec 84.120%
 best acc: 84.120000
 Epoch: [24][0/391] Time 0.150 (0.150) Data 0.122 (0.122) Loss
 0.2889 (0.2889) Prec 91.406% (91.406%)
 Epoch: [24][100/391] Time 0.023 (0.026) Data 0.002 (0.004) Loss
 0.4449 (0.3531) Prec 89.062% (88.127%)
 Epoch: [24][200/391] Time 0.029 (0.026) Data 0.007 (0.004) Loss
 0.4288 (0.3603) Prec 85.156% (87.698%)
 Epoch: [24][300/391] Time 0.025 (0.026) Data 0.003 (0.004) Loss
 0.2085 (0.3604) Prec 92.188% (87.632%)
 Validation starts
 Test: [0/79] Time 0.106 (0.106) Loss 0.5566 (0.5566) Prec 82.812%
 (82.812%)
 * Prec 81.080%
 best acc: 84.120000
 Epoch: [25][0/391] Time 0.157 (0.157) Data 0.121 (0.121) Loss
 0.4013 (0.4013) Prec 84.375% (84.375%)
 Epoch: [25][100/391] Time 0.027 (0.026) Data 0.006 (0.004) Loss
 0.3936 (0.3410) Prec 86.719% (88.506%)
 Epoch: [25][200/391] Time 0.027 (0.025) Data 0.006 (0.004) Loss
 0.3414 (0.3432) Prec 85.938% (88.371%)
 Epoch: [25][300/391] Time 0.024 (0.025) Data 0.002 (0.004) Loss
 0.3839 (0.3457) Prec 88.281% (88.276%)
 Validation starts
 Test: [0/79] Time 0.110 (0.110) Loss 0.5161 (0.5161) Prec 82.812%
 (82.812%)
 * Prec 84.360%
 best acc: 84.360000
 Epoch: [26][0/391] Time 0.155 (0.155) Data 0.126 (0.126) Loss
 0.3351 (0.3351) Prec 89.062% (89.062%)
 Epoch: [26][100/391] Time 0.027 (0.026) Data 0.005 (0.005) Loss
 0.2916 (0.3300) Prec 91.406% (88.769%)
 Epoch: [26][200/391] Time 0.029 (0.026) Data 0.008 (0.004) Loss
 0.3340 (0.3290) Prec 86.719% (88.732%)
 Epoch: [26][300/391] Time 0.029 (0.026) Data 0.007 (0.004) Loss
 0.3396 (0.3299) Prec 88.281% (88.816%)
 Validation starts
 Test: [0/79] Time 0.104 (0.104) Loss 0.3289 (0.3289) Prec 89.062%
 (89.062%)
 * Prec 85.260%

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best acc: 85.260000
Epoch: [27][0/391]      Time 0.148 (0.148)      Data 0.119 (0.119)      Loss
0.2601 (0.2601)      Prec 92.969% (92.969%)
Epoch: [27][100/391]    Time 0.028 (0.026)      Data 0.006 (0.004)      Loss
0.2829 (0.3040)      Prec 92.188% (89.828%)
Epoch: [27][200/391]    Time 0.027 (0.025)      Data 0.006 (0.004)      Loss
0.3569 (0.3081)      Prec 89.844% (89.688%)
Epoch: [27][300/391]    Time 0.031 (0.026)      Data 0.008 (0.004)      Loss
0.3176 (0.3183)      Prec 86.719% (89.369%)
Validation starts
Test: [0/79]      Time 0.106 (0.106)      Loss 0.3917 (0.3917)      Prec 87.500%
(87.500%)
* Prec 84.920%
best acc: 85.260000
Epoch: [28][0/391]      Time 0.162 (0.162)      Data 0.125 (0.125)      Loss
0.2246 (0.2246)      Prec 92.969% (92.969%)
Epoch: [28][100/391]    Time 0.026 (0.026)      Data 0.005 (0.004)      Loss
0.2380 (0.2877)      Prec 91.406% (89.975%)
Epoch: [28][200/391]    Time 0.025 (0.025)      Data 0.004 (0.003)      Loss
0.2980 (0.3025)      Prec 89.844% (89.591%)
Epoch: [28][300/391]    Time 0.031 (0.026)      Data 0.002 (0.003)      Loss
0.2887 (0.3053)      Prec 86.719% (89.587%)
Validation starts
Test: [0/79]      Time 0.105 (0.105)      Loss 0.3935 (0.3935)      Prec 85.156%
(85.156%)
* Prec 84.590%
best acc: 85.260000
Epoch: [29][0/391]      Time 0.157 (0.157)      Data 0.128 (0.128)      Loss
0.3156 (0.3156)      Prec 91.406% (91.406%)
Epoch: [29][100/391]    Time 0.029 (0.027)      Data 0.007 (0.005)      Loss
0.3104 (0.2940)      Prec 92.188% (90.053%)
Epoch: [29][200/391]    Time 0.028 (0.026)      Data 0.006 (0.005)      Loss
0.3622 (0.2969)      Prec 86.719% (89.902%)
Epoch: [29][300/391]    Time 0.028 (0.026)      Data 0.007 (0.004)      Loss
0.2202 (0.2984)      Prec 89.844% (89.800%)
Validation starts
Test: [0/79]      Time 0.103 (0.103)      Loss 0.3967 (0.3967)      Prec 88.281%
(88.281%)
* Prec 85.290%
best acc: 85.290000
Epoch: [30][0/391]      Time 0.150 (0.150)      Data 0.119 (0.119)      Loss
0.3313 (0.3313)      Prec 89.844% (89.844%)
Epoch: [30][100/391]    Time 0.028 (0.026)      Data 0.008 (0.005)      Loss
0.2678 (0.2706)      Prec 92.188% (90.710%)
Epoch: [30][200/391]    Time 0.020 (0.025)      Data 0.000 (0.005)      Loss
0.2449 (0.2854)      Prec 91.406% (90.213%)
Epoch: [30][300/391]    Time 0.026 (0.026)      Data 0.007 (0.004)      Loss
0.4143 (0.2854)      Prec 87.500% (90.173%)

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Validation starts

Test: [0/79] Time 0.098 (0.098) Loss 0.3473 (0.3473) Prec 91.406%
(91.406%)

* Prec 84.990%

best acc: 85.290000

Epoch: [31][0/391] Time 0.145 (0.145) Data 0.118 (0.118) Loss
0.2256 (0.2256) Prec 91.406% (91.406%)

Epoch: [31][100/391] Time 0.026 (0.025) Data 0.007 (0.005) Loss
0.2300 (0.2654) Prec 91.406% (90.780%)

Epoch: [31][200/391] Time 0.025 (0.025) Data 0.005 (0.005) Loss
0.1392 (0.2680) Prec 95.312% (90.940%)

Epoch: [31][300/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.3055 (0.2734) Prec 89.844% (90.648%)

Validation starts

Test: [0/79] Time 0.101 (0.101) Loss 0.3909 (0.3909) Prec 86.719%
(86.719%)

* Prec 85.680%

best acc: 85.680000

Epoch: [32][0/391] Time 0.148 (0.148) Data 0.120 (0.120) Loss
0.3050 (0.3050) Prec 89.844% (89.844%)

Epoch: [32][100/391] Time 0.028 (0.025) Data 0.008 (0.006) Loss
0.2849 (0.2502) Prec 89.062% (91.414%)

Epoch: [32][200/391] Time 0.029 (0.025) Data 0.010 (0.005) Loss
0.2410 (0.2606) Prec 92.188% (91.056%)

Epoch: [32][300/391] Time 0.019 (0.025) Data 0.000 (0.005) Loss
0.3274 (0.2636) Prec 88.281% (90.988%)

Validation starts

Test: [0/79] Time 0.097 (0.097) Loss 0.3635 (0.3635) Prec 88.281%
(88.281%)

* Prec 87.110%

best acc: 87.110000

Epoch: [33][0/391] Time 0.143 (0.143) Data 0.116 (0.116) Loss
0.3047 (0.3047) Prec 85.156% (85.156%)

Epoch: [33][100/391] Time 0.020 (0.025) Data 0.000 (0.006) Loss
0.2406 (0.2496) Prec 95.312% (91.499%)

Epoch: [33][200/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.2314 (0.2562) Prec 92.188% (91.352%)

Epoch: [33][300/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
0.3420 (0.2586) Prec 87.500% (91.240%)

Validation starts

Test: [0/79] Time 0.105 (0.105) Loss 0.3011 (0.3011) Prec 89.844%
(89.844%)

* Prec 85.210%

best acc: 87.110000

Epoch: [34][0/391] Time 0.144 (0.144) Data 0.117 (0.117) Loss
0.2414 (0.2414) Prec 92.188% (92.188%)

Epoch: [34][100/391] Time 0.019 (0.026) Data 0.000 (0.006) Loss
0.4892 (0.2414) Prec 86.719% (91.894%)

Epoch: [34][200/391] Time 0.019 (0.025) Data 0.000 (0.006) Loss
0.3961 (0.2530) Prec 86.719% (91.476%)
Epoch: [34][300/391] Time 0.022 (0.025) Data 0.000 (0.006) Loss
0.2710 (0.2526) Prec 93.750% (91.536%)
Validation starts
Test: [0/79] Time 0.095 (0.095) Loss 0.3853 (0.3853) Prec 87.500%
(87.500%)
* Prec 85.130%
best acc: 87.110000

Epoch: [35][0/391] Time 0.147 (0.147) Data 0.120 (0.120) Loss
0.3034 (0.3034) Prec 89.844% (89.844%)
Epoch: [35][100/391] Time 0.030 (0.026) Data 0.003 (0.006) Loss
0.3316 (0.2374) Prec 89.062% (92.033%)
Epoch: [35][200/391] Time 0.020 (0.028) Data 0.000 (0.005) Loss
0.1313 (0.2453) Prec 96.094% (91.861%)
Epoch: [35][300/391] Time 0.029 (0.027) Data 0.005 (0.005) Loss
0.1782 (0.2455) Prec 91.406% (91.674%)
Validation starts
Test: [0/79] Time 0.097 (0.097) Loss 0.4656 (0.4656) Prec 86.719%
(86.719%)
* Prec 86.330%
best acc: 87.110000

Epoch: [36][0/391] Time 0.144 (0.144) Data 0.118 (0.118) Loss
0.1726 (0.1726) Prec 92.969% (92.969%)
Epoch: [36][100/391] Time 0.021 (0.026) Data 0.000 (0.005) Loss
0.2337 (0.2335) Prec 92.188% (91.909%)
Epoch: [36][200/391] Time 0.027 (0.025) Data 0.008 (0.005) Loss
0.2969 (0.2393) Prec 90.625% (91.873%)
Epoch: [36][300/391] Time 0.027 (0.025) Data 0.001 (0.005) Loss
0.1792 (0.2410) Prec 95.312% (91.899%)
Validation starts
Test: [0/79] Time 0.096 (0.096) Loss 0.3172 (0.3172) Prec 90.625%
(90.625%)
* Prec 86.040%
best acc: 87.110000

Epoch: [37][0/391] Time 0.150 (0.150) Data 0.122 (0.122) Loss
0.1922 (0.1922) Prec 92.969% (92.969%)
Epoch: [37][100/391] Time 0.030 (0.026) Data 0.011 (0.006) Loss
0.2663 (0.2316) Prec 91.406% (91.979%)
Epoch: [37][200/391] Time 0.027 (0.026) Data 0.001 (0.005) Loss
0.1360 (0.2332) Prec 96.094% (92.013%)
Epoch: [37][300/391] Time 0.021 (0.026) Data 0.002 (0.005) Loss
0.2764 (0.2367) Prec 90.625% (91.925%)
Validation starts
Test: [0/79] Time 0.097 (0.097) Loss 0.2579 (0.2579) Prec 92.188%
(92.188%)
* Prec 86.050%
best acc: 87.110000

Epoch: [38][0/391] Time 0.147 (0.147) Data 0.120 (0.120) Loss
0.2782 (0.2782) Prec 91.406% (91.406%)

Epoch: [38][100/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss
0.3353 (0.2176) Prec 90.625% (92.667%)

Epoch: [38][200/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss
0.2169 (0.2223) Prec 92.969% (92.463%)

Epoch: [38][300/391] Time 0.026 (0.025) Data 0.005 (0.005) Loss
0.1556 (0.2295) Prec 96.094% (92.226%)

Validation starts

Test: [0/79] Time 0.099 (0.099) Loss 0.2334 (0.2334) Prec 92.188%
(92.188%)

* Prec 86.380%

best acc: 87.110000

Epoch: [39][0/391] Time 0.151 (0.151) Data 0.116 (0.116) Loss
0.1371 (0.1371) Prec 96.094% (96.094%)

Epoch: [39][100/391] Time 0.022 (0.026) Data 0.002 (0.005) Loss
0.1838 (0.2126) Prec 93.750% (93.162%)

Epoch: [39][200/391] Time 0.027 (0.025) Data 0.008 (0.005) Loss
0.2490 (0.2181) Prec 92.188% (92.685%)

Epoch: [39][300/391] Time 0.043 (0.025) Data 0.012 (0.005) Loss
0.2370 (0.2215) Prec 93.750% (92.582%)

Validation starts

Test: [0/79] Time 0.096 (0.096) Loss 0.3408 (0.3408) Prec 88.281%
(88.281%)

* Prec 86.120%

best acc: 87.110000

Epoch: [40][0/391] Time 0.144 (0.144) Data 0.116 (0.116) Loss
0.1393 (0.1393) Prec 95.312% (95.312%)

Epoch: [40][100/391] Time 0.020 (0.026) Data 0.000 (0.006) Loss
0.2458 (0.2149) Prec 89.062% (92.860%)

Epoch: [40][200/391] Time 0.021 (0.025) Data 0.002 (0.006) Loss
0.2464 (0.2203) Prec 89.844% (92.545%)

Epoch: [40][300/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
0.2118 (0.2178) Prec 92.969% (92.650%)

Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.2867 (0.2867) Prec 90.625%
(90.625%)

* Prec 87.200%

best acc: 87.200000

Epoch: [41][0/391] Time 0.143 (0.143) Data 0.116 (0.116) Loss
0.2327 (0.2327) Prec 90.625% (90.625%)

Epoch: [41][100/391] Time 0.030 (0.025) Data 0.011 (0.006) Loss
0.1267 (0.2008) Prec 96.875% (93.139%)

Epoch: [41][200/391] Time 0.028 (0.025) Data 0.009 (0.005) Loss
0.1409 (0.2051) Prec 96.094% (93.035%)

Epoch: [41][300/391] Time 0.028 (0.025) Data 0.009 (0.005) Loss
0.2703 (0.2060) Prec 92.969% (92.974%)

Validation starts

Test: [0/79] Time 0.105 (0.105) Loss 0.3368 (0.3368) Prec 89.062%
(89.062%)

* Prec 85.680%

best acc: 87.200000

Epoch: [42][0/391] Time 0.152 (0.152) Data 0.116 (0.116) Loss
0.2108 (0.2108) Prec 92.188% (92.188%)

Epoch: [42][100/391] Time 0.026 (0.026) Data 0.007 (0.005) Loss
0.2599 (0.1984) Prec 91.406% (93.162%)

Epoch: [42][200/391] Time 0.027 (0.025) Data 0.008 (0.005) Loss
0.2608 (0.2067) Prec 92.969% (92.809%)

Epoch: [42][300/391] Time 0.027 (0.025) Data 0.008 (0.005) Loss
0.2929 (0.2090) Prec 89.844% (92.790%)

Validation starts

Test: [0/79] Time 0.096 (0.096) Loss 0.3467 (0.3467) Prec 86.719%
(86.719%)

* Prec 87.210%

best acc: 87.210000

Epoch: [43][0/391] Time 0.148 (0.148) Data 0.120 (0.120) Loss
0.1319 (0.1319) Prec 95.312% (95.312%)

Epoch: [43][100/391] Time 0.020 (0.026) Data 0.001 (0.006) Loss
0.1896 (0.1844) Prec 94.531% (93.371%)

Epoch: [43][200/391] Time 0.020 (0.025) Data 0.000 (0.006) Loss
0.1253 (0.1898) Prec 96.094% (93.319%)

Epoch: [43][300/391] Time 0.020 (0.025) Data 0.001 (0.005) Loss
0.2249 (0.1977) Prec 92.188% (93.114%)

Validation starts

Test: [0/79] Time 0.094 (0.094) Loss 0.2533 (0.2533) Prec 93.750%
(93.750%)

* Prec 88.020%

best acc: 88.020000

Epoch: [44][0/391] Time 0.156 (0.156) Data 0.121 (0.121) Loss
0.1725 (0.1725) Prec 94.531% (94.531%)

Epoch: [44][100/391] Time 0.022 (0.026) Data 0.001 (0.005) Loss
0.0971 (0.1724) Prec 96.094% (94.052%)

Epoch: [44][200/391] Time 0.026 (0.025) Data 0.006 (0.005) Loss
0.2710 (0.1825) Prec 92.188% (93.715%)

Epoch: [44][300/391] Time 0.025 (0.025) Data 0.005 (0.005) Loss
0.1893 (0.1852) Prec 95.312% (93.659%)

Validation starts

Test: [0/79] Time 0.100 (0.100) Loss 0.1997 (0.1997) Prec 92.188%
(92.188%)

* Prec 86.690%

best acc: 88.020000

Epoch: [45][0/391] Time 0.145 (0.145) Data 0.117 (0.117) Loss
0.0940 (0.0940) Prec 97.656% (97.656%)

Epoch: [45][100/391] Time 0.027 (0.025) Data 0.008 (0.005) Loss
0.2243 (0.1780) Prec 91.406% (93.905%)

Epoch: [45][200/391] Time 0.021 (0.024) Data 0.002 (0.005) Loss

0.2466 (0.1904) Prec 91.406% (93.548%)
Epoch: [45][300/391] Time 0.020 (0.024) Data 0.000 (0.005) Loss
0.1527 (0.1939) Prec 93.750% (93.340%)
Validation starts
Test: [0/79] Time 0.096 (0.096) Loss 0.3200 (0.3200) Prec 91.406%
(91.406%)
* Prec 86.520%
best acc: 88.020000
Epoch: [46][0/391] Time 0.146 (0.146) Data 0.119 (0.119) Loss
0.1345 (0.1345) Prec 96.094% (96.094%)
Epoch: [46][100/391] Time 0.027 (0.025) Data 0.008 (0.005) Loss
0.1958 (0.1819) Prec 92.188% (93.889%)
Epoch: [46][200/391] Time 0.030 (0.025) Data 0.009 (0.005) Loss
0.2090 (0.1812) Prec 92.969% (93.797%)
Epoch: [46][300/391] Time 0.032 (0.025) Data 0.012 (0.005) Loss
0.2699 (0.1853) Prec 89.062% (93.618%)
Validation starts
Test: [0/79] Time 0.101 (0.101) Loss 0.3583 (0.3583) Prec 89.844%
(89.844%)
* Prec 86.840%
best acc: 88.020000
Epoch: [47][0/391] Time 0.146 (0.146) Data 0.118 (0.118) Loss
0.2351 (0.2351) Prec 91.406% (91.406%)
Epoch: [47][100/391] Time 0.028 (0.025) Data 0.008 (0.006) Loss
0.1256 (0.1718) Prec 96.094% (94.175%)
Epoch: [47][200/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
0.2411 (0.1756) Prec 92.188% (94.053%)
Epoch: [47][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.3444 (0.1750) Prec 89.844% (94.090%)
Validation starts
Test: [0/79] Time 0.095 (0.095) Loss 0.3985 (0.3985) Prec 85.938%
(85.938%)
* Prec 85.900%
best acc: 88.020000
Epoch: [48][0/391] Time 0.146 (0.146) Data 0.118 (0.118) Loss
0.2046 (0.2046) Prec 92.969% (92.969%)
Epoch: [48][100/391] Time 0.028 (0.026) Data 0.008 (0.006) Loss
0.1821 (0.1793) Prec 92.969% (93.704%)
Epoch: [48][200/391] Time 0.021 (0.025) Data 0.000 (0.005) Loss
0.1993 (0.1776) Prec 92.969% (93.820%)
Epoch: [48][300/391] Time 0.021 (0.025) Data 0.001 (0.005) Loss
0.1742 (0.1759) Prec 94.531% (93.919%)
Validation starts
Test: [0/79] Time 0.098 (0.098) Loss 0.4988 (0.4988) Prec 87.500%
(87.500%)
* Prec 86.680%
best acc: 88.020000
Epoch: [49][0/391] Time 0.154 (0.154) Data 0.118 (0.118) Loss

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0.1563 (0.1563)    Prec 93.750% (93.750%)
Epoch: [49][100/391]    Time 0.028 (0.026)    Data 0.008 (0.005)    Loss
0.1742 (0.1570)    Prec 93.750% (94.454%)
Epoch: [49][200/391]    Time 0.028 (0.025)    Data 0.007 (0.005)    Loss
0.1263 (0.1679)    Prec 96.875% (94.181%)
Epoch: [49][300/391]    Time 0.028 (0.025)    Data 0.008 (0.005)    Loss
0.1526 (0.1699)    Prec 95.312% (94.194%)
Validation starts
Test: [0/79]    Time 0.102 (0.102)    Loss 0.3529 (0.3529)    Prec 90.625%
(90.625%)
* Prec 87.550%
best acc: 88.020000
Epoch: [50][0/391]    Time 0.155 (0.155)    Data 0.124 (0.124)    Loss
0.1376 (0.1376)    Prec 94.531% (94.531%)
Epoch: [50][100/391]    Time 0.026 (0.025)    Data 0.006 (0.005)    Loss
0.1348 (0.1626)    Prec 94.531% (94.493%)
Epoch: [50][200/391]    Time 0.020 (0.025)    Data 0.000 (0.005)    Loss
0.1744 (0.1650)    Prec 92.969% (94.450%)
Epoch: [50][300/391]    Time 0.020 (0.025)    Data 0.001 (0.005)    Loss
0.1889 (0.1690)    Prec 94.531% (94.324%)
Validation starts
Test: [0/79]    Time 0.095 (0.095)    Loss 0.4079 (0.4079)    Prec 85.938%
(85.938%)
* Prec 87.430%
best acc: 88.020000
Epoch: [51][0/391]    Time 0.154 (0.154)    Data 0.125 (0.125)    Loss
0.1039 (0.1039)    Prec 96.094% (96.094%)
Epoch: [51][100/391]    Time 0.026 (0.026)    Data 0.006 (0.006)    Loss
0.2347 (0.1550)    Prec 89.844% (94.686%)
Epoch: [51][200/391]    Time 0.021 (0.025)    Data 0.002 (0.005)    Loss
0.3719 (0.1636)    Prec 88.281% (94.403%)
Epoch: [51][300/391]    Time 0.019 (0.025)    Data 0.000 (0.005)    Loss
0.2663 (0.1708)    Prec 90.625% (94.108%)
Validation starts
Test: [0/79]    Time 0.096 (0.096)    Loss 0.3192 (0.3192)    Prec 91.406%
(91.406%)
* Prec 87.580%
best acc: 88.020000
Epoch: [52][0/391]    Time 0.142 (0.142)    Data 0.115 (0.115)    Loss
0.2066 (0.2066)    Prec 92.969% (92.969%)
Epoch: [52][100/391]    Time 0.020 (0.026)    Data 0.000 (0.006)    Loss
0.1490 (0.1483)    Prec 95.312% (94.848%)
Epoch: [52][200/391]    Time 0.020 (0.025)    Data 0.000 (0.006)    Loss
0.1898 (0.1556)    Prec 95.312% (94.574%)
Epoch: [52][300/391]    Time 0.020 (0.025)    Data 0.000 (0.005)    Loss
0.1860 (0.1615)    Prec 94.531% (94.391%)
Validation starts
Test: [0/79]    Time 0.098 (0.098)    Loss 0.2872 (0.2872)    Prec 93.750%

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(93.750%)

* Prec 88.340%

best acc: 88.340000

Epoch: [53] [0/391]	Time 0.153 (0.153)	Data 0.117 (0.117)	Loss
0.0622 (0.0622)	Prec 98.438% (98.438%)		
Epoch: [53] [100/391]	Time 0.021 (0.026)	Data 0.001 (0.005)	Loss
0.1244 (0.1451)	Prec 96.875% (94.995%)		
Epoch: [53] [200/391]	Time 0.026 (0.025)	Data 0.007 (0.005)	Loss
0.1271 (0.1520)	Prec 95.312% (94.792%)		
Epoch: [53] [300/391]	Time 0.020 (0.025)	Data 0.000 (0.005)	Loss
0.1925 (0.1530)	Prec 93.750% (94.791%)		

Validation starts

Test: [0/79] Time 0.097 (0.097) Loss 0.3184 (0.3184) Prec 89.062% (89.062%)

* Prec 87.880%

best acc: 88.340000

Epoch: [54] [0/391]	Time 0.146 (0.146)	Data 0.118 (0.118)	Loss
0.1134 (0.1134)	Prec 96.875% (96.875%)		
Epoch: [54] [100/391]	Time 0.026 (0.027)	Data 0.007 (0.005)	Loss
0.3014 (0.1476)	Prec 90.625% (95.073%)		
Epoch: [54] [200/391]	Time 0.029 (0.026)	Data 0.010 (0.005)	Loss
0.1811 (0.1470)	Prec 93.750% (94.943%)		
Epoch: [54] [300/391]	Time 0.021 (0.025)	Data 0.002 (0.005)	Loss
0.0986 (0.1540)	Prec 96.875% (94.757%)		

Validation starts

Test: [0/79] Time 0.096 (0.096) Loss 0.2604 (0.2604) Prec 92.969% (92.969%)

* Prec 87.420%

best acc: 88.340000

Epoch: [55] [0/391]	Time 0.158 (0.158)	Data 0.122 (0.122)	Loss
0.0996 (0.0996)	Prec 96.094% (96.094%)		
Epoch: [55] [100/391]	Time 0.021 (0.026)	Data 0.002 (0.005)	Loss
0.1355 (0.1560)	Prec 95.312% (94.663%)		
Epoch: [55] [200/391]	Time 0.020 (0.025)	Data 0.001 (0.005)	Loss
0.1568 (0.1620)	Prec 92.188% (94.395%)		
Epoch: [55] [300/391]	Time 0.020 (0.025)	Data 0.000 (0.005)	Loss
0.2369 (0.1582)	Prec 90.625% (94.573%)		

Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.3461 (0.3461) Prec 89.062% (89.062%)

* Prec 88.030%

best acc: 88.340000

Epoch: [56] [0/391]	Time 0.147 (0.147)	Data 0.120 (0.120)	Loss
0.1591 (0.1591)	Prec 92.969% (92.969%)		
Epoch: [56] [100/391]	Time 0.029 (0.026)	Data 0.009 (0.006)	Loss
0.1636 (0.1330)	Prec 92.188% (95.459%)		
Epoch: [56] [200/391]	Time 0.030 (0.026)	Data 0.010 (0.006)	Loss
0.1277 (0.1387)	Prec 95.312% (95.250%)		

Epoch: [56][300/391] Time 0.030 (0.025) Data 0.011 (0.006) Loss
0.1897 (0.1431) Prec 92.969% (95.061%)
Validation starts
Test: [0/79] Time 0.093 (0.093) Loss 0.4002 (0.4002) Prec 87.500%
(87.500%)
* Prec 88.070%
best acc: 88.340000

Epoch: [57][0/391] Time 0.144 (0.144) Data 0.116 (0.116) Loss
0.1500 (0.1500) Prec 95.312% (95.312%)
Epoch: [57][100/391] Time 0.027 (0.025) Data 0.004 (0.005) Loss
0.0976 (0.1281) Prec 96.875% (95.653%)
Epoch: [57][200/391] Time 0.020 (0.025) Data 0.001 (0.005) Loss
0.1944 (0.1397) Prec 94.531% (95.285%)
Epoch: [57][300/391] Time 0.020 (0.025) Data 0.001 (0.005) Loss
0.2098 (0.1427) Prec 92.969% (95.123%)
Validation starts
Test: [0/79] Time 0.097 (0.097) Loss 0.4708 (0.4708) Prec 86.719%
(86.719%)
* Prec 87.510%
best acc: 88.340000

Epoch: [58][0/391] Time 0.144 (0.144) Data 0.116 (0.116) Loss
0.1036 (0.1036) Prec 97.656% (97.656%)
Epoch: [58][100/391] Time 0.030 (0.025) Data 0.010 (0.005) Loss
0.1749 (0.1390) Prec 92.969% (95.220%)
Epoch: [58][200/391] Time 0.026 (0.025) Data 0.006 (0.005) Loss
0.0945 (0.1392) Prec 96.094% (95.149%)
Epoch: [58][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.1250 (0.1420) Prec 95.312% (95.092%)
Validation starts
Test: [0/79] Time 0.098 (0.098) Loss 0.3009 (0.3009) Prec 93.750%
(93.750%)
* Prec 87.680%
best acc: 88.340000

Epoch: [59][0/391] Time 0.154 (0.154) Data 0.116 (0.116) Loss
0.1044 (0.1044) Prec 97.656% (97.656%)
Epoch: [59][100/391] Time 0.021 (0.025) Data 0.000 (0.005) Loss
0.1778 (0.1287) Prec 94.531% (95.521%)
Epoch: [59][200/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss
0.1191 (0.1350) Prec 94.531% (95.305%)
Epoch: [59][300/391] Time 0.028 (0.025) Data 0.009 (0.005) Loss
0.1721 (0.1418) Prec 96.094% (95.043%)
Validation starts
Test: [0/79] Time 0.095 (0.095) Loss 0.3302 (0.3302) Prec 89.062%
(89.062%)
* Prec 88.230%
best acc: 88.340000

Epoch: [60][0/391] Time 0.147 (0.147) Data 0.120 (0.120) Loss
0.1325 (0.1325) Prec 95.312% (95.312%)

Epoch: [60][100/391] Time 0.024 (0.027) Data 0.000 (0.005) Loss
 0.1219 (0.1011) Prec 96.094% (96.620%)
 Epoch: [60][200/391] Time 0.020 (0.026) Data 0.000 (0.004) Loss
 0.0639 (0.0916) Prec 98.438% (96.894%)
 Epoch: [60][300/391] Time 0.020 (0.025) Data 0.000 (0.004) Loss
 0.1051 (0.0872) Prec 96.094% (97.033%)
 Validation starts
 Test: [0/79] Time 0.093 (0.093) Loss 0.2579 (0.2579) Prec 92.188%
 (92.188%)
 * Prec 90.830%
 best acc: 90.830000
 Epoch: [61][0/391] Time 0.144 (0.144) Data 0.116 (0.116) Loss
 0.0813 (0.0813) Prec 96.094% (96.094%)
 Epoch: [61][100/391] Time 0.030 (0.025) Data 0.011 (0.006) Loss
 0.0546 (0.0597) Prec 97.656% (98.004%)
 Epoch: [61][200/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
 0.0788 (0.0583) Prec 96.094% (98.041%)
 Epoch: [61][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
 0.0180 (0.0566) Prec 100.000% (98.147%)
 Validation starts
 Test: [0/79] Time 0.098 (0.098) Loss 0.2659 (0.2659) Prec 92.188%
 (92.188%)
 * Prec 90.730%
 best acc: 90.830000
 Epoch: [62][0/391] Time 0.148 (0.148) Data 0.120 (0.120) Loss
 0.0256 (0.0256) Prec 100.000% (100.000%)
 Epoch: [62][100/391] Time 0.019 (0.026) Data 0.000 (0.006) Loss
 0.0316 (0.0478) Prec 98.438% (98.499%)
 Epoch: [62][200/391] Time 0.019 (0.026) Data 0.000 (0.006) Loss
 0.0719 (0.0483) Prec 96.875% (98.465%)
 Epoch: [62][300/391] Time 0.019 (0.025) Data 0.000 (0.005) Loss
 0.0493 (0.0500) Prec 98.438% (98.380%)
 Validation starts
 Test: [0/79] Time 0.099 (0.099) Loss 0.3020 (0.3020) Prec 92.188%
 (92.188%)
 * Prec 90.860%
 best acc: 90.860000
 Epoch: [63][0/391] Time 0.153 (0.153) Data 0.118 (0.118) Loss
 0.0383 (0.0383) Prec 99.219% (99.219%)
 Epoch: [63][100/391] Time 0.027 (0.026) Data 0.004 (0.006) Loss
 0.0492 (0.0424) Prec 98.438% (98.577%)
 Epoch: [63][200/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
 0.0220 (0.0449) Prec 99.219% (98.507%)
 Epoch: [63][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
 0.0411 (0.0445) Prec 98.438% (98.526%)
 Validation starts
 Test: [0/79] Time 0.099 (0.099) Loss 0.2833 (0.2833) Prec 92.969%
 (92.969%)

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* Prec 90.970%
best acc: 90.970000
Epoch: [64][0/391]      Time 0.146 (0.146)      Data 0.118 (0.118)      Loss
0.0616 (0.0616)      Prec 95.312% (95.312%)
Epoch: [64][100/391]    Time 0.031 (0.026)      Data 0.012 (0.007)      Loss
0.0384 (0.0410)      Prec 98.438% (98.523%)
Epoch: [64][200/391]    Time 0.027 (0.026)      Data 0.006 (0.006)      Loss
0.0933 (0.0394)      Prec 96.875% (98.616%)
Epoch: [64][300/391]    Time 0.030 (0.025)      Data 0.011 (0.006)      Loss
0.0137 (0.0399)      Prec 100.000% (98.617%)
Validation starts
Test: [0/79]      Time 0.100 (0.100)      Loss 0.2343 (0.2343)      Prec 93.750%
(93.750%)
* Prec 90.910%
best acc: 90.970000
Epoch: [65][0/391]      Time 0.155 (0.155)      Data 0.120 (0.120)      Loss
0.0100 (0.0100)      Prec 100.000% (100.000%)
Epoch: [65][100/391]    Time 0.029 (0.026)      Data 0.010 (0.005)      Loss
0.0094 (0.0364)      Prec 100.000% (98.809%)
Epoch: [65][200/391]    Time 0.029 (0.025)      Data 0.009 (0.005)      Loss
0.0703 (0.0376)      Prec 99.219% (98.725%)
Epoch: [65][300/391]    Time 0.029 (0.025)      Data 0.009 (0.005)      Loss
0.0504 (0.0376)      Prec 98.438% (98.752%)
Validation starts
Test: [0/79]      Time 0.101 (0.101)      Loss 0.2581 (0.2581)      Prec 92.969%
(92.969%)
* Prec 91.130%
best acc: 91.130000
Epoch: [66][0/391]      Time 0.146 (0.146)      Data 0.118 (0.118)      Loss
0.0479 (0.0479)      Prec 98.438% (98.438%)
Epoch: [66][100/391]    Time 0.020 (0.025)      Data 0.000 (0.006)      Loss
0.0864 (0.0358)      Prec 96.094% (98.817%)
Epoch: [66][200/391]    Time 0.027 (0.025)      Data 0.007 (0.005)      Loss
0.0063 (0.0358)      Prec 100.000% (98.760%)
Epoch: [66][300/391]    Time 0.021 (0.025)      Data 0.002 (0.005)      Loss
0.0181 (0.0355)      Prec 99.219% (98.785%)
Validation starts
Test: [0/79]      Time 0.095 (0.095)      Loss 0.2953 (0.2953)      Prec 93.750%
(93.750%)
* Prec 91.110%
best acc: 91.130000
Epoch: [67][0/391]      Time 0.146 (0.146)      Data 0.119 (0.119)      Loss
0.0068 (0.0068)      Prec 100.000% (100.000%)
Epoch: [67][100/391]    Time 0.026 (0.025)      Data 0.006 (0.005)      Loss
0.0356 (0.0326)      Prec 99.219% (98.948%)
Epoch: [67][200/391]    Time 0.031 (0.025)      Data 0.010 (0.005)      Loss
0.0194 (0.0336)      Prec 99.219% (98.865%)
Epoch: [67][300/391]    Time 0.029 (0.025)      Data 0.010 (0.005)      Loss

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0.0129 (0.0340) Prec 100.000% (98.824%)
Validation starts
Test: [0/79] Time 0.098 (0.098) Loss 0.2573 (0.2573) Prec 90.625%
(90.625%)
* Prec 90.960%
best acc: 91.130000
Epoch: [68][0/391] Time 0.156 (0.156) Data 0.121 (0.121) Loss
0.0384 (0.0384) Prec 99.219% (99.219%)
Epoch: [68][100/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
0.0934 (0.0309) Prec 97.656% (99.002%)
Epoch: [68][200/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.0784 (0.0300) Prec 96.875% (98.966%)
Epoch: [68][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.0323 (0.0313) Prec 99.219% (98.902%)
Validation starts
Test: [0/79] Time 0.097 (0.097) Loss 0.2978 (0.2978) Prec 91.406%
(91.406%)
* Prec 91.120%
best acc: 91.130000
Epoch: [69][0/391] Time 0.145 (0.145) Data 0.118 (0.118) Loss
0.0488 (0.0488) Prec 98.438% (98.438%)
Epoch: [69][100/391] Time 0.022 (0.025) Data 0.003 (0.005) Loss
0.0173 (0.0313) Prec 99.219% (98.871%)
Epoch: [69][200/391] Time 0.025 (0.025) Data 0.005 (0.005) Loss
0.0265 (0.0314) Prec 99.219% (98.916%)
Epoch: [69][300/391] Time 0.019 (0.025) Data 0.000 (0.005) Loss
0.0029 (0.0298) Prec 100.000% (99.006%)
Validation starts
Test: [0/79] Time 0.100 (0.100) Loss 0.2411 (0.2411) Prec 92.188%
(92.188%)
* Prec 91.090%
best acc: 91.130000
Epoch: [70][0/391] Time 0.143 (0.143) Data 0.116 (0.116) Loss
0.0140 (0.0140) Prec 98.438% (98.438%)
Epoch: [70][100/391] Time 0.023 (0.025) Data 0.003 (0.006) Loss
0.0098 (0.0249) Prec 100.000% (99.196%)
Epoch: [70][200/391] Time 0.020 (0.025) Data 0.000 (0.006) Loss
0.0270 (0.0271) Prec 98.438% (99.063%)
Epoch: [70][300/391] Time 0.029 (0.025) Data 0.003 (0.005) Loss
0.0360 (0.0278) Prec 98.438% (99.019%)
Validation starts
Test: [0/79] Time 0.098 (0.098) Loss 0.2044 (0.2044) Prec 93.750%
(93.750%)
* Prec 91.150%
best acc: 91.150000
Epoch: [71][0/391] Time 0.149 (0.149) Data 0.114 (0.114) Loss
0.0198 (0.0198) Prec 99.219% (99.219%)
Epoch: [71][100/391] Time 0.027 (0.028) Data 0.007 (0.005) Loss

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0.0313 (0.0325)    Prec 98.438% (98.925%)
Epoch: [71][200/391]    Time 0.029 (0.026)    Data 0.010 (0.005)    Loss
0.0170 (0.0284)    Prec 99.219% (99.087%)
Epoch: [71][300/391]    Time 0.032 (0.026)    Data 0.012 (0.005)    Loss
0.0486 (0.0271)    Prec 98.438% (99.099%)
Validation starts
Test: [0/79]    Time 0.096 (0.096)    Loss 0.2662 (0.2662)    Prec 91.406%
(91.406%)
* Prec 91.170%
best acc: 91.170000
Epoch: [72][0/391]    Time 0.147 (0.147)    Data 0.120 (0.120)    Loss
0.0130 (0.0130)    Prec 99.219% (99.219%)
Epoch: [72][100/391]    Time 0.030 (0.026)    Data 0.011 (0.007)    Loss
0.0250 (0.0287)    Prec 98.438% (98.948%)
Epoch: [72][200/391]    Time 0.020 (0.026)    Data 0.000 (0.006)    Loss
0.0209 (0.0273)    Prec 99.219% (99.036%)
Epoch: [72][300/391]    Time 0.020 (0.026)    Data 0.001 (0.006)    Loss
0.0049 (0.0276)    Prec 100.000% (99.055%)
Validation starts
Test: [0/79]    Time 0.091 (0.091)    Loss 0.3403 (0.3403)    Prec 91.406%
(91.406%)
* Prec 91.160%
best acc: 91.170000
Epoch: [73][0/391]    Time 0.150 (0.150)    Data 0.123 (0.123)    Loss
0.0179 (0.0179)    Prec 99.219% (99.219%)
Epoch: [73][100/391]    Time 0.029 (0.025)    Data 0.009 (0.006)    Loss
0.0113 (0.0218)    Prec 100.000% (99.250%)
Epoch: [73][200/391]    Time 0.030 (0.025)    Data 0.009 (0.005)    Loss
0.0122 (0.0228)    Prec 99.219% (99.207%)
Epoch: [73][300/391]    Time 0.028 (0.025)    Data 0.008 (0.005)    Loss
0.0658 (0.0239)    Prec 98.438% (99.188%)
Validation starts
Test: [0/79]    Time 0.100 (0.100)    Loss 0.2953 (0.2953)    Prec 89.062%
(89.062%)
* Prec 90.920%
best acc: 91.170000
Epoch: [74][0/391]    Time 0.151 (0.151)    Data 0.123 (0.123)    Loss
0.0203 (0.0203)    Prec 99.219% (99.219%)
Epoch: [74][100/391]    Time 0.029 (0.026)    Data 0.010 (0.006)    Loss
0.0159 (0.0252)    Prec 99.219% (99.103%)
Epoch: [74][200/391]    Time 0.029 (0.025)    Data 0.009 (0.006)    Loss
0.0221 (0.0244)    Prec 99.219% (99.137%)
Epoch: [74][300/391]    Time 0.022 (0.025)    Data 0.002 (0.005)    Loss
0.0050 (0.0237)    Prec 100.000% (99.154%)
Validation starts
Test: [0/79]    Time 0.099 (0.099)    Loss 0.2147 (0.2147)    Prec 92.188%
(92.188%)
* Prec 91.170%

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best acc: 91.170000
Epoch: [75][0/391]      Time 0.146 (0.146)      Data 0.118 (0.118)      Loss
0.0206 (0.0206)      Prec 99.219% (99.219%)
Epoch: [75][100/391]    Time 0.030 (0.026)      Data 0.011 (0.006)      Loss
0.0152 (0.0241)      Prec 100.000% (99.157%)
Epoch: [75][200/391]    Time 0.029 (0.025)      Data 0.009 (0.006)      Loss
0.0047 (0.0253)      Prec 100.000% (99.122%)
Epoch: [75][300/391]    Time 0.033 (0.025)      Data 0.010 (0.006)      Loss
0.0126 (0.0246)      Prec 99.219% (99.154%)
Validation starts
Test: [0/79]      Time 0.097 (0.097)      Loss 0.2195 (0.2195)      Prec 94.531%
(94.531%)
* Prec 91.010%
best acc: 91.170000
Epoch: [76][0/391]      Time 0.145 (0.145)      Data 0.117 (0.117)      Loss
0.0020 (0.0020)      Prec 100.000% (100.000%)
Epoch: [76][100/391]    Time 0.021 (0.025)      Data 0.002 (0.005)      Loss
0.0086 (0.0243)      Prec 99.219% (99.234%)
Epoch: [76][200/391]    Time 0.031 (0.025)      Data 0.009 (0.005)      Loss
0.0417 (0.0226)      Prec 98.438% (99.265%)
Epoch: [76][300/391]    Time 0.027 (0.025)      Data 0.008 (0.005)      Loss
0.0040 (0.0223)      Prec 100.000% (99.273%)
Validation starts
Test: [0/79]      Time 0.099 (0.099)      Loss 0.2264 (0.2264)      Prec 94.531%
(94.531%)
* Prec 90.980%
best acc: 91.170000
Epoch: [77][0/391]      Time 0.145 (0.145)      Data 0.118 (0.118)      Loss
0.0190 (0.0190)      Prec 99.219% (99.219%)
Epoch: [77][100/391]    Time 0.028 (0.026)      Data 0.008 (0.006)      Loss
0.0415 (0.0216)      Prec 98.438% (99.172%)
Epoch: [77][200/391]    Time 0.031 (0.025)      Data 0.011 (0.005)      Loss
0.0167 (0.0221)      Prec 99.219% (99.195%)
Epoch: [77][300/391]    Time 0.030 (0.025)      Data 0.009 (0.005)      Loss
0.0059 (0.0227)      Prec 100.000% (99.206%)
Validation starts
Test: [0/79]      Time 0.101 (0.101)      Loss 0.2414 (0.2414)      Prec 92.969%
(92.969%)
* Prec 91.310%
best acc: 91.310000
Epoch: [78][0/391]      Time 0.143 (0.143)      Data 0.115 (0.115)      Loss
0.0144 (0.0144)      Prec 99.219% (99.219%)
Epoch: [78][100/391]    Time 0.028 (0.025)      Data 0.009 (0.006)      Loss
0.0099 (0.0215)      Prec 99.219% (99.211%)
Epoch: [78][200/391]    Time 0.029 (0.025)      Data 0.009 (0.005)      Loss
0.0064 (0.0204)      Prec 100.000% (99.269%)
Epoch: [78][300/391]    Time 0.028 (0.025)      Data 0.002 (0.005)      Loss
0.0238 (0.0202)      Prec 99.219% (99.289%)

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Validation starts

Test: [0/79] Time 0.095 (0.095) Loss 0.2557 (0.2557) Prec 93.750%
(93.750%)

* Prec 91.020%

best acc: 91.310000

Epoch: [79] [0/391]	Time 0.145 (0.145)	Data 0.118 (0.118)	Loss
0.0265 (0.0265)	Prec 99.219% (99.219%)		
Epoch: [79] [100/391]	Time 0.029 (0.026)	Data 0.009 (0.006)	Loss
0.0133 (0.0178)	Prec 100.000% (99.420%)		
Epoch: [79] [200/391]	Time 0.029 (0.025)	Data 0.010 (0.005)	Loss
0.0212 (0.0198)	Prec 99.219% (99.335%)		
Epoch: [79] [300/391]	Time 0.030 (0.025)	Data 0.010 (0.005)	Loss
0.0186 (0.0194)	Prec 98.438% (99.351%)		

Validation starts

Test: [0/79] Time 0.095 (0.095) Loss 0.2760 (0.2760) Prec 92.969%
(92.969%)

* Prec 91.110%

best acc: 91.310000

Epoch: [80] [0/391]	Time 0.158 (0.158)	Data 0.123 (0.123)	Loss
0.0066 (0.0066)	Prec 100.000% (100.000%)		
Epoch: [80] [100/391]	Time 0.031 (0.026)	Data 0.011 (0.006)	Loss
0.0178 (0.0201)	Prec 100.000% (99.366%)		
Epoch: [80] [200/391]	Time 0.030 (0.026)	Data 0.010 (0.006)	Loss
0.0040 (0.0194)	Prec 100.000% (99.398%)		
Epoch: [80] [300/391]	Time 0.031 (0.025)	Data 0.011 (0.005)	Loss
0.0463 (0.0192)	Prec 97.656% (99.359%)		

Validation starts

Test: [0/79] Time 0.094 (0.094) Loss 0.2124 (0.2124) Prec 96.094%
(96.094%)

* Prec 91.230%

best acc: 91.310000

Epoch: [81] [0/391]	Time 0.142 (0.142)	Data 0.114 (0.114)	Loss
0.0058 (0.0058)	Prec 100.000% (100.000%)		
Epoch: [81] [100/391]	Time 0.020 (0.025)	Data 0.000 (0.005)	Loss
0.0782 (0.0188)	Prec 97.656% (99.428%)		
Epoch: [81] [200/391]	Time 0.024 (0.025)	Data 0.005 (0.005)	Loss
0.0234 (0.0196)	Prec 99.219% (99.370%)		
Epoch: [81] [300/391]	Time 0.020 (0.024)	Data 0.000 (0.005)	Loss
0.0075 (0.0187)	Prec 100.000% (99.372%)		

Validation starts

Test: [0/79] Time 0.104 (0.104) Loss 0.3025 (0.3025) Prec 93.750%
(93.750%)

* Prec 91.220%

best acc: 91.310000

Epoch: [82] [0/391]	Time 0.153 (0.153)	Data 0.125 (0.125)	Loss
0.0119 (0.0119)	Prec 99.219% (99.219%)		
Epoch: [82] [100/391]	Time 0.022 (0.025)	Data 0.002 (0.005)	Loss
0.0074 (0.0172)	Prec 100.000% (99.435%)		

Epoch: [82][200/391] Time 0.020 (0.024) Data 0.000 (0.005) Loss
 0.0104 (0.0186) Prec 99.219% (99.394%)
 Epoch: [82][300/391] Time 0.022 (0.024) Data 0.002 (0.004) Loss
 0.0160 (0.0182) Prec 99.219% (99.393%)
 Validation starts
 Test: [0/79] Time 0.104 (0.104) Loss 0.2939 (0.2939) Prec 92.969%
 (92.969%)
 * Prec 91.350%
 best acc: 91.350000
 Epoch: [83][0/391] Time 0.152 (0.152) Data 0.116 (0.116) Loss
 0.0181 (0.0181) Prec 99.219% (99.219%)
 Epoch: [83][100/391] Time 0.028 (0.026) Data 0.008 (0.006) Loss
 0.0040 (0.0205) Prec 100.000% (99.319%)
 Epoch: [83][200/391] Time 0.029 (0.025) Data 0.010 (0.006) Loss
 0.0048 (0.0184) Prec 100.000% (99.374%)
 Epoch: [83][300/391] Time 0.031 (0.025) Data 0.012 (0.006) Loss
 0.0055 (0.0186) Prec 100.000% (99.359%)
 Validation starts
 Test: [0/79] Time 0.099 (0.099) Loss 0.2994 (0.2994) Prec 92.188%
 (92.188%)
 * Prec 91.200%
 best acc: 91.350000
 Epoch: [84][0/391] Time 0.145 (0.145) Data 0.118 (0.118) Loss
 0.0076 (0.0076) Prec 100.000% (100.000%)
 Epoch: [84][100/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss
 0.0064 (0.0136) Prec 100.000% (99.582%)
 Epoch: [84][200/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
 0.0336 (0.0138) Prec 99.219% (99.572%)
 Epoch: [84][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
 0.0015 (0.0144) Prec 100.000% (99.567%)
 Validation starts
 Test: [0/79] Time 0.097 (0.097) Loss 0.2385 (0.2385) Prec 93.750%
 (93.750%)
 * Prec 91.180%
 best acc: 91.350000
 Epoch: [85][0/391] Time 0.146 (0.146) Data 0.119 (0.119) Loss
 0.0158 (0.0158) Prec 99.219% (99.219%)
 Epoch: [85][100/391] Time 0.029 (0.026) Data 0.010 (0.007) Loss
 0.1031 (0.0175) Prec 97.656% (99.412%)
 Epoch: [85][200/391] Time 0.029 (0.025) Data 0.010 (0.006) Loss
 0.0185 (0.0174) Prec 99.219% (99.386%)
 Epoch: [85][300/391] Time 0.030 (0.025) Data 0.010 (0.006) Loss
 0.0247 (0.0168) Prec 99.219% (99.406%)
 Validation starts
 Test: [0/79] Time 0.102 (0.102) Loss 0.2431 (0.2431) Prec 92.969%
 (92.969%)
 * Prec 91.220%
 best acc: 91.350000

Epoch: [86][0/391] Time 0.144 (0.144) Data 0.117 (0.117) Loss
0.0094 (0.0094) Prec 99.219% (99.219%)
Epoch: [86][100/391] Time 0.028 (0.025) Data 0.008 (0.006) Loss
0.0042 (0.0159) Prec 100.000% (99.443%)
Epoch: [86][200/391] Time 0.026 (0.025) Data 0.007 (0.005) Loss
0.0386 (0.0156) Prec 99.219% (99.483%)
Epoch: [86][300/391] Time 0.030 (0.025) Data 0.011 (0.005) Loss
0.0088 (0.0155) Prec 100.000% (99.489%)
Validation starts
Test: [0/79] Time 0.096 (0.096) Loss 0.2191 (0.2191) Prec 94.531%
(94.531%)
* Prec 91.250%
best acc: 91.350000
Epoch: [87][0/391] Time 0.144 (0.144) Data 0.117 (0.117) Loss
0.0275 (0.0275) Prec 98.438% (98.438%)
Epoch: [87][100/391] Time 0.025 (0.025) Data 0.006 (0.005) Loss
0.0053 (0.0154) Prec 100.000% (99.489%)
Epoch: [87][200/391] Time 0.030 (0.025) Data 0.010 (0.005) Loss
0.0379 (0.0154) Prec 99.219% (99.468%)
Epoch: [87][300/391] Time 0.030 (0.025) Data 0.010 (0.005) Loss
0.0257 (0.0158) Prec 99.219% (99.439%)
Validation starts
Test: [0/79] Time 0.095 (0.095) Loss 0.2392 (0.2392) Prec 94.531%
(94.531%)
* Prec 91.350%
best acc: 91.350000
Epoch: [88][0/391] Time 0.151 (0.151) Data 0.116 (0.116) Loss
0.0074 (0.0074) Prec 100.000% (100.000%)
Epoch: [88][100/391] Time 0.029 (0.025) Data 0.010 (0.005) Loss
0.0040 (0.0138) Prec 100.000% (99.520%)
Epoch: [88][200/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
0.0346 (0.0153) Prec 99.219% (99.491%)
Epoch: [88][300/391] Time 0.028 (0.024) Data 0.008 (0.005) Loss
0.0088 (0.0148) Prec 100.000% (99.515%)
Validation starts
Test: [0/79] Time 0.099 (0.099) Loss 0.2821 (0.2821) Prec 94.531%
(94.531%)
* Prec 91.220%
best acc: 91.350000
Epoch: [89][0/391] Time 0.150 (0.150) Data 0.120 (0.120) Loss
0.0177 (0.0177) Prec 99.219% (99.219%)
Epoch: [89][100/391] Time 0.030 (0.027) Data 0.010 (0.006) Loss
0.0081 (0.0135) Prec 100.000% (99.497%)
Epoch: [89][200/391] Time 0.029 (0.026) Data 0.009 (0.006) Loss
0.0051 (0.0148) Prec 100.000% (99.452%)
Epoch: [89][300/391] Time 0.029 (0.025) Data 0.010 (0.006) Loss
0.0129 (0.0150) Prec 99.219% (99.463%)
Validation starts

Test: [0/79] Time 0.096 (0.096) Loss 0.2607 (0.2607) Prec 92.969%
(92.969%)

* Prec 91.240%

best acc: 91.350000

Epoch: [90][0/391] Time 0.147 (0.147) Data 0.119 (0.119) Loss
0.0209 (0.0209) Prec 98.438% (98.438%)

Epoch: [90][100/391] Time 0.025 (0.026) Data 0.005 (0.005) Loss
0.0070 (0.0162) Prec 100.000% (99.451%)

Epoch: [90][200/391] Time 0.030 (0.026) Data 0.010 (0.005) Loss
0.0320 (0.0161) Prec 98.438% (99.440%)

Epoch: [90][300/391] Time 0.030 (0.025) Data 0.011 (0.005) Loss
0.0058 (0.0152) Prec 100.000% (99.478%)

Validation starts

Test: [0/79] Time 0.100 (0.100) Loss 0.2710 (0.2710) Prec 95.312%
(95.312%)

* Prec 91.320%

best acc: 91.350000

Epoch: [91][0/391] Time 0.157 (0.157) Data 0.128 (0.128) Loss
0.0192 (0.0192) Prec 100.000% (100.000%)

Epoch: [91][100/391] Time 0.027 (0.026) Data 0.007 (0.006) Loss
0.0022 (0.0127) Prec 100.000% (99.644%)

Epoch: [91][200/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.0094 (0.0130) Prec 100.000% (99.623%)

Epoch: [91][300/391] Time 0.020 (0.025) Data 0.001 (0.005) Loss
0.0023 (0.0137) Prec 100.000% (99.585%)

Validation starts

Test: [0/79] Time 0.096 (0.096) Loss 0.3044 (0.3044) Prec 93.750%
(93.750%)

* Prec 91.170%

best acc: 91.350000

Epoch: [92][0/391] Time 0.157 (0.157) Data 0.122 (0.122) Loss
0.0026 (0.0026) Prec 100.000% (100.000%)

Epoch: [92][100/391] Time 0.021 (0.027) Data 0.002 (0.005) Loss
0.0014 (0.0149) Prec 100.000% (99.505%)

Epoch: [92][200/391] Time 0.021 (0.026) Data 0.002 (0.005) Loss
0.0339 (0.0146) Prec 98.438% (99.491%)

Epoch: [92][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.0395 (0.0152) Prec 98.438% (99.478%)

Validation starts

Test: [0/79] Time 0.094 (0.094) Loss 0.2314 (0.2314) Prec 93.750%
(93.750%)

* Prec 91.290%

best acc: 91.350000

Epoch: [93][0/391] Time 0.151 (0.151) Data 0.116 (0.116) Loss
0.0309 (0.0309) Prec 99.219% (99.219%)

Epoch: [93][100/391] Time 0.020 (0.027) Data 0.000 (0.005) Loss
0.0321 (0.0127) Prec 98.438% (99.582%)

Epoch: [93][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss

0.0013 (0.0139) Prec 100.000% (99.541%)
 Epoch: [93][300/391] Time 0.023 (0.026) Data 0.000 (0.005) Loss
 0.0038 (0.0143) Prec 100.000% (99.520%)
 Validation starts
 Test: [0/79] Time 0.101 (0.101) Loss 0.2518 (0.2518) Prec 95.312%
 (95.312%)
 * Prec 91.140%
 best acc: 91.350000
 Epoch: [94][0/391] Time 0.153 (0.153) Data 0.118 (0.118) Loss
 0.0089 (0.0089) Prec 100.000% (100.000%)
 Epoch: [94][100/391] Time 0.022 (0.026) Data 0.002 (0.005) Loss
 0.0014 (0.0158) Prec 100.000% (99.482%)
 Epoch: [94][200/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
 0.0192 (0.0139) Prec 99.219% (99.553%)
 Epoch: [94][300/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
 0.0117 (0.0143) Prec 99.219% (99.528%)
 Validation starts
 Test: [0/79] Time 0.101 (0.101) Loss 0.1968 (0.1968) Prec 96.094%
 (96.094%)
 * Prec 91.260%
 best acc: 91.350000
 Epoch: [95][0/391] Time 0.145 (0.145) Data 0.117 (0.117) Loss
 0.0137 (0.0137) Prec 99.219% (99.219%)
 Epoch: [95][100/391] Time 0.027 (0.027) Data 0.008 (0.005) Loss
 0.0072 (0.0161) Prec 100.000% (99.474%)
 Epoch: [95][200/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
 0.0143 (0.0148) Prec 100.000% (99.495%)
 Epoch: [95][300/391] Time 0.027 (0.025) Data 0.008 (0.005) Loss
 0.0108 (0.0146) Prec 99.219% (99.502%)
 Validation starts
 Test: [0/79] Time 0.096 (0.096) Loss 0.2699 (0.2699) Prec 93.750%
 (93.750%)
 * Prec 91.230%
 best acc: 91.350000
 Epoch: [96][0/391] Time 0.146 (0.146) Data 0.118 (0.118) Loss
 0.0180 (0.0180) Prec 99.219% (99.219%)
 Epoch: [96][100/391] Time 0.024 (0.026) Data 0.003 (0.005) Loss
 0.0032 (0.0135) Prec 100.000% (99.606%)
 Epoch: [96][200/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss
 0.0217 (0.0136) Prec 99.219% (99.588%)
 Epoch: [96][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
 0.0070 (0.0142) Prec 100.000% (99.530%)
 Validation starts
 Test: [0/79] Time 0.100 (0.100) Loss 0.2311 (0.2311) Prec 94.531%
 (94.531%)
 * Prec 91.330%
 best acc: 91.350000
 Epoch: [97][0/391] Time 0.146 (0.146) Data 0.118 (0.118) Loss

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0.0035 (0.0035)    Prec 100.000% (100.000%)
Epoch: [97][100/391]    Time 0.021 (0.026)    Data 0.000 (0.005)    Loss
0.0028 (0.0121)    Prec 100.000% (99.621%)
Epoch: [97][200/391]    Time 0.020 (0.025)    Data 0.000 (0.005)    Loss
0.0051 (0.0131)    Prec 100.000% (99.592%)
Epoch: [97][300/391]    Time 0.020 (0.025)    Data 0.000 (0.005)    Loss
0.0081 (0.0129)    Prec 100.000% (99.595%)
Validation starts
Test: [0/79]    Time 0.100 (0.100)    Loss 0.2315 (0.2315)    Prec 94.531%
(94.531%)
* Prec 91.360%
best acc: 91.360000
Epoch: [98][0/391]    Time 0.151 (0.151)    Data 0.116 (0.116)    Loss
0.0024 (0.0024)    Prec 100.000% (100.000%)
Epoch: [98][100/391]    Time 0.020 (0.027)    Data 0.000 (0.006)    Loss
0.0062 (0.0145)    Prec 100.000% (99.567%)
Epoch: [98][200/391]    Time 0.021 (0.026)    Data 0.001 (0.005)    Loss
0.0090 (0.0148)    Prec 99.219% (99.534%)
Epoch: [98][300/391]    Time 0.020 (0.026)    Data 0.000 (0.005)    Loss
0.0102 (0.0148)    Prec 99.219% (99.530%)
Validation starts
Test: [0/79]    Time 0.103 (0.103)    Loss 0.2552 (0.2552)    Prec 94.531%
(94.531%)
* Prec 91.180%
best acc: 91.360000
Epoch: [99][0/391]    Time 0.144 (0.144)    Data 0.117 (0.117)    Loss
0.0021 (0.0021)    Prec 100.000% (100.000%)
Epoch: [99][100/391]    Time 0.020 (0.026)    Data 0.000 (0.005)    Loss
0.0176 (0.0135)    Prec 99.219% (99.582%)
Epoch: [99][200/391]    Time 0.021 (0.025)    Data 0.002 (0.005)    Loss
0.0141 (0.0134)    Prec 99.219% (99.553%)
Epoch: [99][300/391]    Time 0.029 (0.026)    Data 0.008 (0.005)    Loss
0.0135 (0.0136)    Prec 100.000% (99.533%)
Validation starts
Test: [0/79]    Time 0.097 (0.097)    Loss 0.2224 (0.2224)    Prec 93.750%
(93.750%)
* Prec 91.200%
best acc: 91.360000

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```

[3]: PATH = "/content/drive/MyDrive/ece284/result/model_best.pth.tar"
model_name = "VGG16_project"
model = VGG16_project()
checkpoint = torch.load(PATH)
model.load_state_dict(checkpoint['state_dict'])
device = torch.device("cuda")

model.cuda()

```

```

model.eval()

test_loss = 0
correct = 0

with torch.no_grad():
    for data, target in testloader:
        data, target = data.to(device), target.to(device) # loading to GPU
        output = model(data)
        pred = output.argmax(dim=1, keepdim=True)
        correct += pred.eq(target.view_as(pred)).sum().item()

test_loss /= len(testloader.dataset)

print('\nTest set: Accuracy: {}/{} ({:.0f}%) \n'.format(
    correct, len(testloader.dataset),
    100. * correct / len(testloader.dataset)))

```

Test set: Accuracy: 9140/10000 (91%)

```

[4]: class SaveOutput:
    def __init__(self):
        self.outputs = []
    def __call__(self, module, module_in):
        self.outputs.append(module_in)
    def clear(self):
        self.outputs = []

##### Save inputs from selected layer #####
save_output = SaveOutput()
i = 0
counter = 0
for layer in model.modules():
    i+=1
    if isinstance(layer, QuantConv2d):
        print(i, "-th layer prehooked")
        counter+=1
        print(layer, counter)
        layer.register_forward_pre_hook(save_output)
#####

dataiter = iter(testloader)
images, labels = next(dataiter)
images = images.to(device)
out = model(images)

```

```

3 -th layer prehooked
QuantConv2d(
  3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 1
7 -th layer prehooked
QuantConv2d(
  64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 2
12 -th layer prehooked
QuantConv2d(
  64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 3
16 -th layer prehooked
QuantConv2d(
  128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 4
21 -th layer prehooked
QuantConv2d(
  128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 5
25 -th layer prehooked
QuantConv2d(
  256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 6
29 -th layer prehooked
QuantConv2d(
  256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 7
34 -th layer prehooked
QuantConv2d(
  256, 8, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 8
38 -th layer prehooked
QuantConv2d(
  8, 8, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 9
41 -th layer prehooked
QuantConv2d(
  8, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False

```

```

    (weight_quant): weight_quantize_fn()
) 10
46 -th layer prehooked
QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 11
50 -th layer prehooked
QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 12
54 -th layer prehooked
QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 13

```

```

[5]: w_bit = 4
weight_q = model.features[27].weight_q # quantized value is stored during the
    ↪ training
w_alpha = model.features[27].weight_quant.wgt_alpha # alpha is defined in
    ↪ your model already. bring it out here
w_delta = w_alpha/(2**(w_bit-1)-1) # delta can be calculated by using alpha
    ↪ and w_bit
weight_int = weight_q/w_delta # w_int can be calculated by weight_q and w_delta
# print(weight_int) # you should see clean integer numbers
print(f"Weight Int Shape: {weight_int.shape}")

```

Weight Int Shape: torch.Size([8, 8, 3, 3])

```

[10]: x_bit = 4
x = save_output.outputs[8][0] # input of the 2nd conv layer
print(f"Input Shape: {x.shape}")
x_alpha = model.features[27].act_alpha
x_delta = x_alpha/(2**x_bit-1)

act_quant_fn = act_quantization(x_bit) # define the quantization function
x_q = act_quant_fn(x, x_alpha) # create the quantized value for x

x_int = x_q/x_delta
print(f"Input Int Sample: {x_int[0,0,0,:5]}")

```

Input Shape: torch.Size([128, 8, 4, 4])
Input Int Sample: tensor([3.0000, 0.0000, 0.0000, 0.0000], device='cuda:0',
grad_fn=<SelectBackward0>)

```
[11]: ##### input floating number / weight quantized version

conv_ref = torch.nn.Conv2d(in_channels = 8, out_channels=8, kernel_size = 3,
    ↪padding=1, bias = False)
conv_ref.weight = torch.nn.parameter.Parameter(weight_int)
output_int = conv_ref(x_int)

output_recovered = output_int*x_delta*w_delta
relu = nn.ReLU(inplace=True)
relu_output_recovered = relu(output_recovered)

output_ref = save_output.outputs[9][0]
```

```
[12]: difference = abs( output_ref - relu_output_recovered )
print(difference.mean())  ## It should be small, e.g., 2.3 in my trained model
```

```
tensor(2.8219e-07, device='cuda:0', grad_fn=<MeanBackward0>)
```

```
[46]: x_pad = torch.zeros(128, 8, 6, 6).to(x_int.device)
x_pad[:, :, 1:5, 1:5] = x_int
X = x_pad[0]
X = X.reshape(X.size(0), -1)
X.size()
```

```
[46]: torch.Size([8, 36])
```

```
[47]: tile_id = 0
nij = 200 # just a random number
#X = a_tile[tile_id,:,nij:nij+64] # [tile_num, array row num, time_steps]

bit_precision = 4
file = open('activation.txt', 'w') #write to file
file.write('#time0row7[msb-lsb],time0row6[msb-lst],...,time0row0[msb-lst]#\n')
file.write('#time1row7[msb-lsb],time1row6[msb-lst],...,time1row0[msb-lst]#\n')
file.write('#.....#\n')

for i in range(X.size(1)): # time step
    for j in range(X.size(0)): # row #
        X_bin = '{0:04b}'.format(int(X[7-j,i].item()+0.001))
        for k in range(bit_precision):
            file.write(X_bin[k])
            #file.write(' ') # for visibility with blank between words, you can use
        file.write('\n')
file.close() #close file
```

```
[48]: print(weight_int.size())
W = torch.reshape(weight_int, (weight_int.size(0), weight_int.size(1), -1))
```

```
print(W.size())
```

```
torch.Size([8, 8, 3, 3])
```

```
torch.Size([8, 8, 9])
```

```
[49]: bit_precision = 4
file = open('weight.txt', 'w') #write to file
file.write('#col0row7[msb-lsb],col0row6[msb-lst],...,col0row0[msb-lst]#\n')
file.write('#col1row7[msb-lsb],col1row6[msb-lst],...,col1row0[msb-lst]#\n')
file.write('#.....#\n')
for kij in range(9):
    for i in range(W.size(0)):
        for j in range(W.size(1)):
            if (W[i, 7-j, kij].item()<0):
                W_bin = '{0:04b}'.format(int(W[i,7-j,kij].
↪item()+2**bit_precision+0.001))
            else:
                W_bin = '{0:04b}'.format(int(W[i,7-j,kij].item()+0.001))
            for k in range(bit_precision):
                file.write(W_bin[k])
            #file.write(' ') # for visibility with blank between words, you
↪can use
            file.write('\n')
file.close() #close file
```

```
[50]: W[0,:,0]
```

```
[50]: tensor([-5.0000, -0.0000,  2.0000,  7.0000,  7.0000,  3.0000, -3.0000,  2.0000],
        device='cuda:0', grad_fn=<SelectBackward0>)
```

```
[51]: p_nijg = range(X.size(1)) ## psum nij group
psum = torch.zeros(8, len(p_nijg), 9).cuda()
for kij in range(9):
    for nij in p_nijg:          # time domain, sequentially given input
        m = nn.Linear(8, 8, bias=False)
        m.weight = torch.nn.Parameter(W[:, :, kij])
        psum[:, nij, kij] = m(X[:, :, nij]).cuda()
bit_precision = 16
file = open('psum.txt', 'w') #write to file
file.write('#time0col7[msb-lsb],time0col6[msb-lst],...,time0col0[msb-lst]#\n')
file.write('#time1col7[msb-lsb],time1col6[msb-lst],...,time1col0[msb-lst]#\n')
file.write('#.....#\n')
for kij in range(9):
    for i in range(psum.size(1)):
        for j in range(psum.size(0)):
            if (psum[7-j,i,kij].item()<0):
```



```

        P_bin = '{0:016b}'.format(int(psum[7-j,i,kij].
↪item()+2**bit_precision+0.001))
    else:
        P_bin = '{0:016b}'.format(int(psum[7-j,i,kij].item()+0.001))
    for k in range(bit_precision):
        file.write(P_bin[k])
        #file.write(' ') # for visibility with blank between words, you
↪can use
        file.write('\n')
file.close()

```

```

[58]: out = relu(output_int[0])
out = torch.reshape(out, (out.size(0), -1))
bit_precision = 16
file = open('output.txt', 'w') #write to file
file.write('#time0col7[msb-lsb],time0col6[msb-lst],...,time0col0[msb-lst]#\n')
file.write('#time1col7[msb-lsb],time1col6[msb-lst],...,time1col0[msb-lst]#\n')
file.write('#.....#\n')

for i in range(out.size(1)):
    for j in range(out.size(0)):
        if (out[7-j,i].item())<0:
            O_bin = '{0:016b}'.format(int(out[7-j,i].item()+2**bit_precision+0.
↪001))
        else:
            O_bin = '{0:016b}'.format(int(out[7-j,i].item()+0.001))
        for k in range(bit_precision):
            file.write(O_bin[k])
            #file.write(' ') # for visibility with blank between words, you can use
        file.write('\n')
file.close()

```